

CO1 ASSESSMENT TOOL 1: -CASE STUDY ANALYSIS

AUTONOMOUS ELECTRIC VEHICLE CHARGING NETWORK MANAGEMENT SYSTEM

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1. Problem Understanding

Rapid growth in electric vehicle (EV) adoption requires continuous, real-time coordination of charging stations. Manual and isolated management leads to long driver wait times, uneven load distribution on the power grid, frequent station downtime, and inefficient use of charging infrastructure.

Introduction:

- Electric mobility is central to reducing carbon emissions and building sustainable transportation systems.
- However, existing charging networks face several challenges, including unpredictable demand, limited grid capacity during peak hours, station downtime, poor interoperability between networks, and static, non-dynamic pricing.
- These challenges create range anxiety among drivers and make it difficult for operators to scale charging infrastructure while keeping the grid stable.

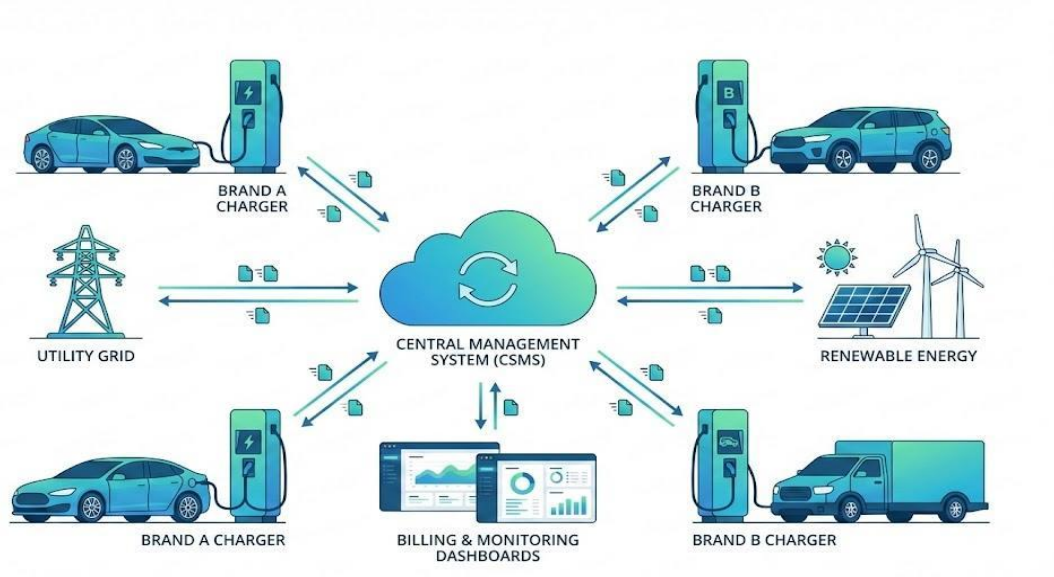
Objectives:

- Monitor charging station status and availability in real time
- Optimize load distribution across the grid
- Predict demand and forecast station availability
- Automate dynamic pricing and slot booking
- Generate usage and maintenance reports
- Notify drivers and operators of station status and faults

2. Stakeholder Analysis

Stakeholders: EV Owners/Drivers, Charging Station Operators, Grid/Utility Providers, Fleet Managers, System Administrators, Payment Gateway Providers, Government/Regulatory Bodies.

Roles and challenges are explained through station monitoring, load balancing, billing accuracy, regulatory compliance, uptime assurance and driver experience.



3. Current System Analysis

Existing systems rely on isolated station apps, manual dispatch and fixed, static pricing.

Strengths: simple to deploy, low integration overhead.

Weaknesses: no real-time load balancing, frequent station downtime, poor interoperability between charging networks, static pricing, limited fault detection and delayed maintenance response.

4. Proposed Software Solution

Modules:

- Real-Time Station Monitoring
- Dynamic Load Balancing and Grid Integration

- AI-Based Demand Prediction and Slot Booking
- Automated Billing and Dynamic Pricing
- Fault Detection and Maintenance Alerts
- Reporting and Analytics Dashboard

5. Requirements

- **Functional:** User login/authentication, real-time station availability map, slot booking and reservation, automated billing, load balancing across stations, fault and maintenance alerts, usage reports.
- **Non-functional:** Security, Reliability, Scalability, Maintainability, Performance, Usability, Interoperability.

6. Software Engineering Principles

Modularity, Scalability, Maintainability, Reliability, Security and Usability are applied using a microservices architecture, cloud-based deployment, encrypted authentication, redundant backups and a responsive driver mobile app and operator dashboard.

7. SDLC Justification

Agile Scrum is selected because charging infrastructure requirements, grid regulations and EV technology evolve rapidly. Short sprints, continuous integration and testing, stakeholder feedback and incremental delivery reduce project risk and allow the system to adapt to changing grid and regulatory conditions.

8. Expected Outcomes

Reduced driver wait times, optimized grid load distribution, higher charging station uptime, improved energy efficiency, increased EV adoption and better operator profitability.

9. Risks and Limitations

- Grid capacity constraints
- Network and connectivity dependency

- High implementation and hardware cost
- Cybersecurity risks to payment data and charging infrastructure
- Interoperability issues across different charging standards

10. Future Enhancements

- Vehicle-to-Grid (V2G) integration, blockchain-based billing, AI-driven predictive maintenance, renewable energy source integration and autonomous robotic charging.
- Implement pricing based on demand, time, and usage to optimize revenue.
- Allow EVs to send power back to the grid when needed.
- suggest best charging stations based on traffic conditions

Conclusion

The Autonomous Electric Vehicle Charging Network Management System combines IoT, AI and cloud computing to optimize charging infrastructure, reduce strain on the power grid and enhance the EV ownership experience while following Software Engineering principles.

Code of Conduct:

I V. Pavithra 192524274 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.